NEURAL NETWORKS · GROUP 3

Fake News Detection and LSTM Length Sensitivity

June 2026 · Section 1 Topic 11 · Section 2 Topic 21

Task 1: Fake News Detection

Task 2: LSTM Memory Horizon

OVERVIEW

Two tasks, one theme: when does architecture matter?

Both tasks challenge the assumption that a better model solves everything.

TASK 1

Can we detect fake news from text alone?

LIAR dataset · 12,836 PolitiFact statements

Four models compared head-to-head

Metric: F1 on the FAKE class

Section 1 · Topic 11

TASK 2

How far back can an LSTM remember?

Synthetic delayed-recall task

Vary distance between signal and readout

Metric: binary classification accuracy

Section 2 · Topic 21

TASK 1

17-word statements, borderline labels, no easy signal

12,836

PolitiFact statements (LIAR dataset)

~17

Average words per statement

2-class

Binary: FAKE vs REAL

WHAT MAKES IT HARD

- Short text - limited semantic content per sample
- Borderline labels cluster near the decision boundary (barely-true, half-true)
- Ground truth requires fact-checking knowledge, not just language patterns
- Speaker reputation leaks into labels - content signal is noisy

EVALUATION METRIC

Precision & Recall on FAKE class - used as a triage tool

Four models, one fair comparison

MLP

TF-IDF + linguistic

Bag-of-words features, hand-crafted linguistic signals, shallow network

BiLSTM

Raw text sequences

Bidirectional recurrence over token embeddings, text only

DistilBERT

Pretrained transformer

Fine-tuned on LIAR; 66M params; contextual embeddings from pretraining

Hybrid

Text + metadata

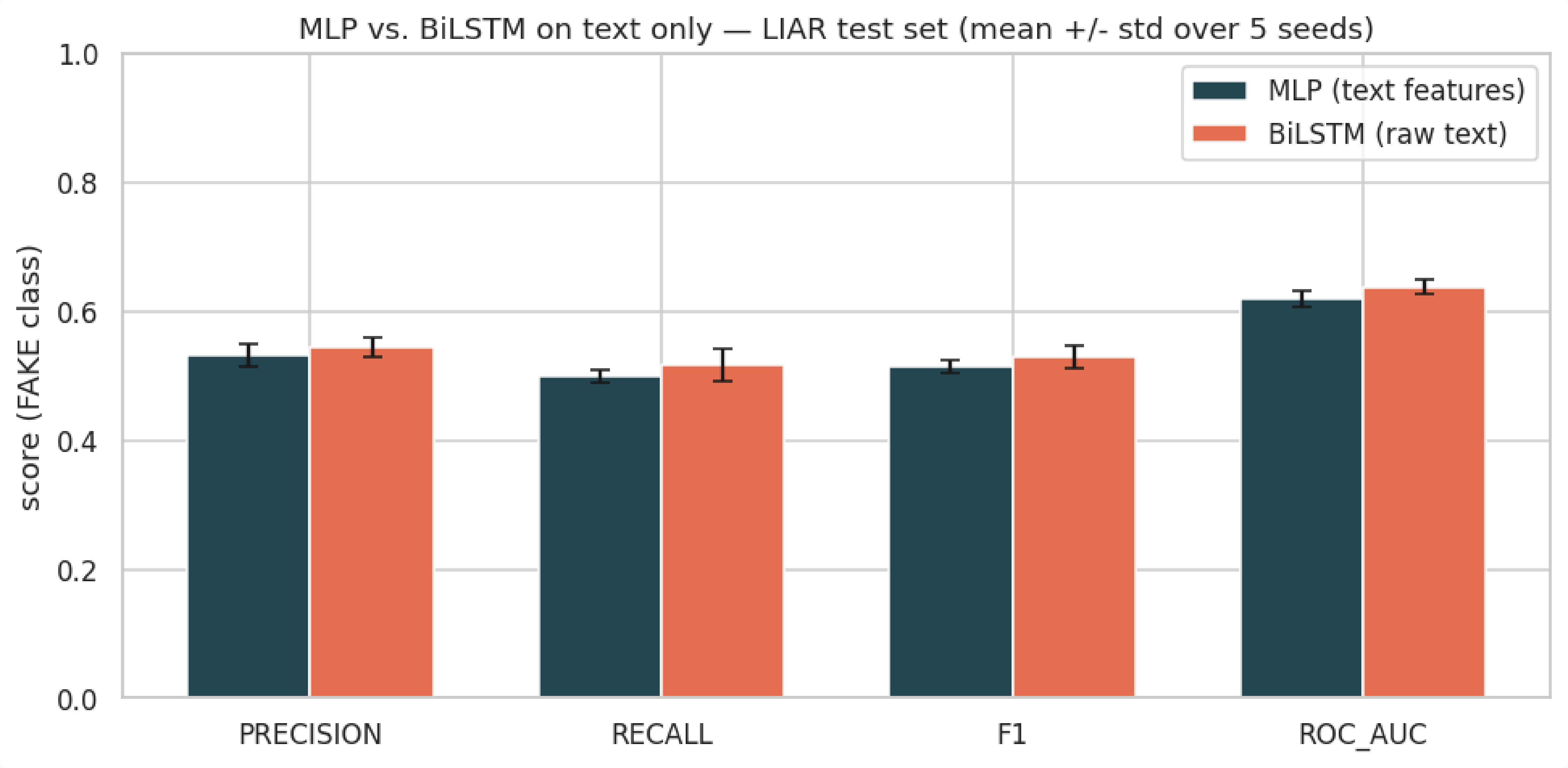
Combines text head with speaker metadata (party, state, history)

KEY DESIGN CHOICE

Text-only head-to-head kept fair - metadata studied separately to isolate what the language signal contributes

TASK 1 · EXPERIMENT 1

On text alone, LSTM machinery buys nothing



0.514
MLP F1
TF-IDF + linguistic features

0.530
BiLSTM F1
Raw text sequences

MCNEMAR'S TEST
p = 0.69 - statistical noise, not a real difference

12 metadata numbers outscore every text model

FEATURE SET	F1 (FAKE)	AUC
Text features only	~0.51	-
Text + metadata (combined)	0.56	-
Metadata only (12 features)	**0.64**	**0.75**

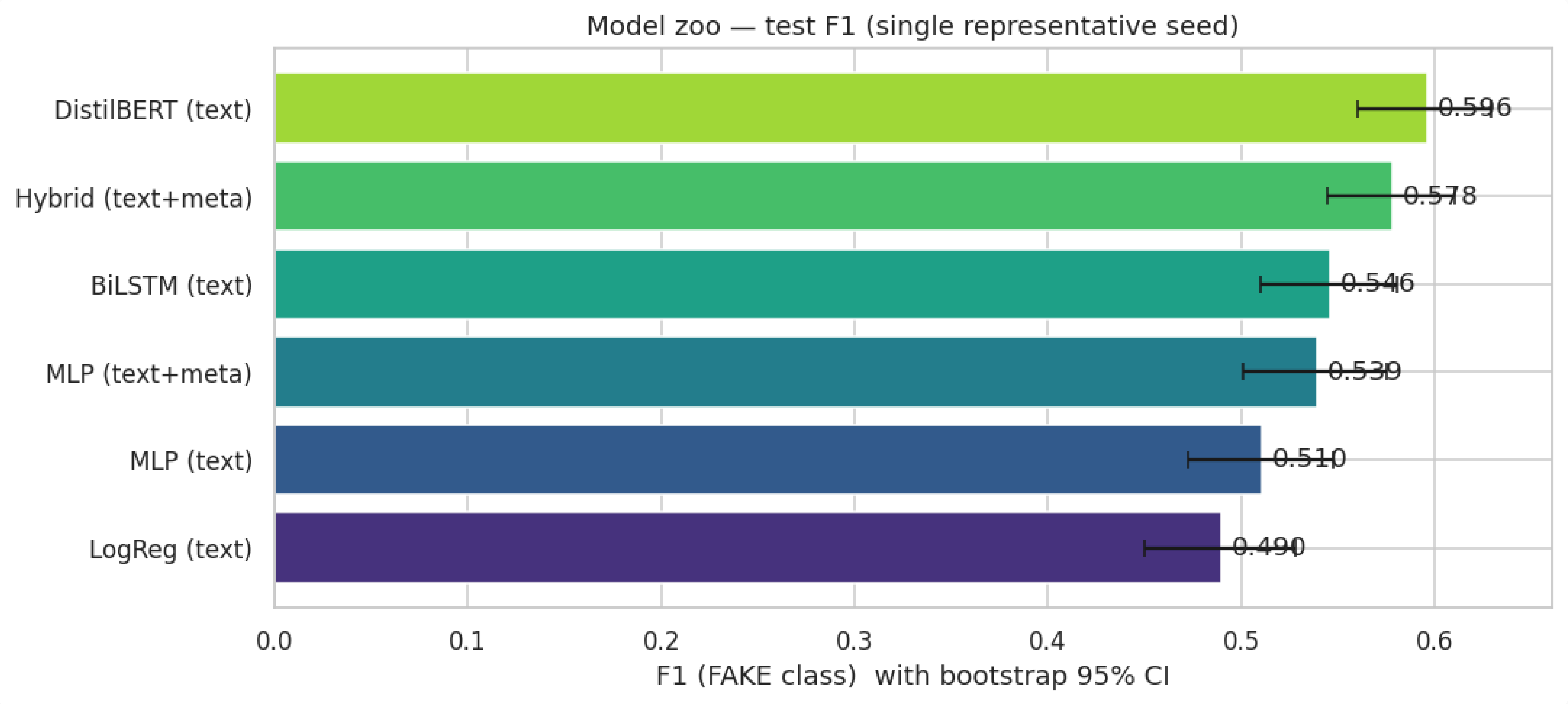
+25%

F1 gain from metadata alone
vs. best text-only model

INSIGHT

LIAR is partially predicting speaker reputation, not content. The label correlates with who said it, not just what was said.

Pretraining is the only real gain on text



0.60 F1

DistilBERT

66M params · pretrained on large corpus

0.51 - 0.53

Scratch models

~1M params · trained on LIAR only

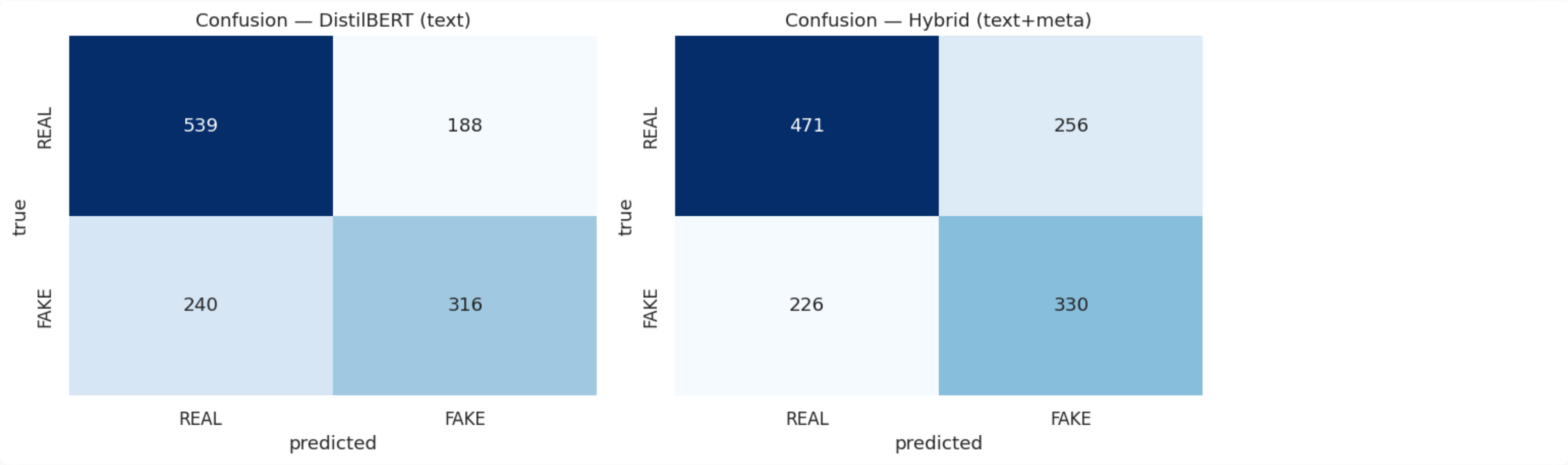
CEILING

The constraint is knowledge, not architecture. Bigger scratch models don't close the gap.

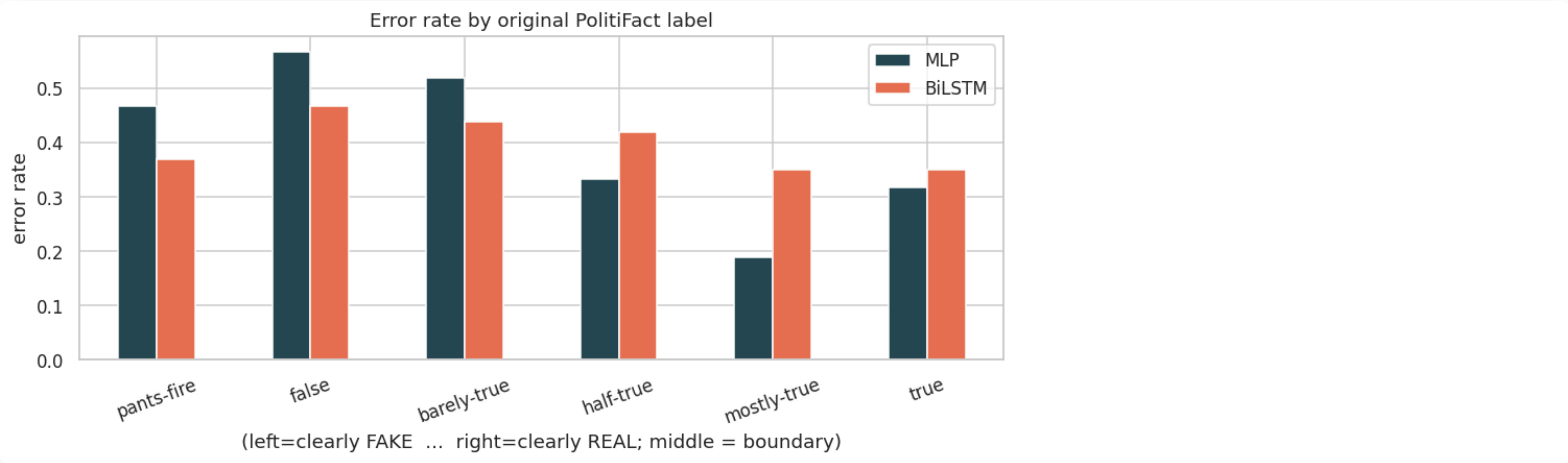
TASK 1 · FAILURE ANALYSIS

Every model misses borderline claims - that is the floor

Confusion Matrices (all models)



Error rate by PolitiFact label



ERROR CLUSTER

Errors concentrate on false / barely-true - labels nearest the binary boundary

EASY END

Mostly-true statements are almost never misclassified across all models

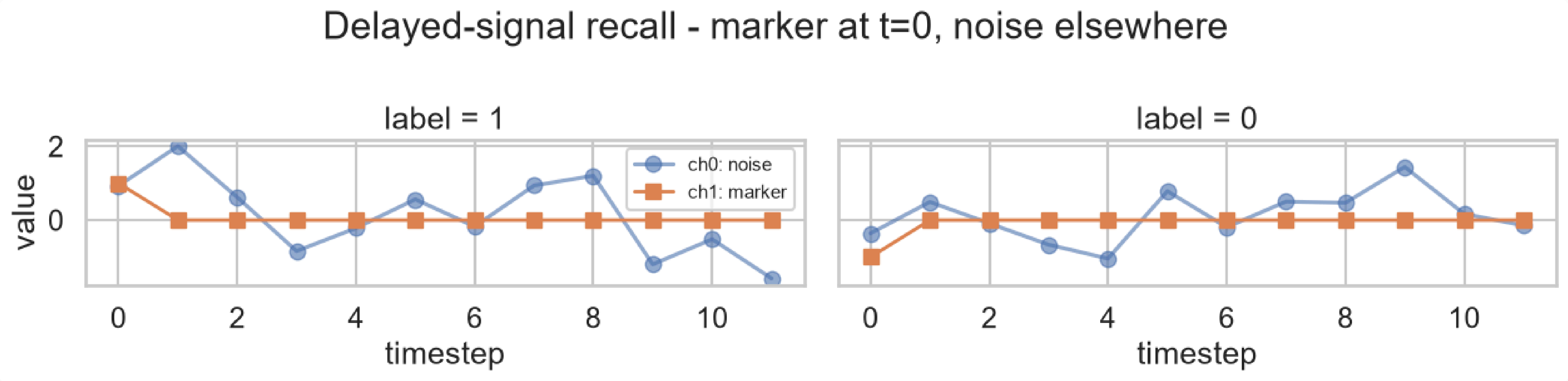
HARD FLOOR

~25% of test set is unsolvable from 18 words alone - fact-checking knowledge required

TASK 2

Is it length, or how far the signal must travel?

Sample sequences - clean marker embedded in noise, readout at end



TASK DESIGN

Synthetic delayed-signal recall: single clean marker buried in random noise, model classifies from the final timestep

ONLY VARIABLE

Distance between marker and readout - everything else held constant

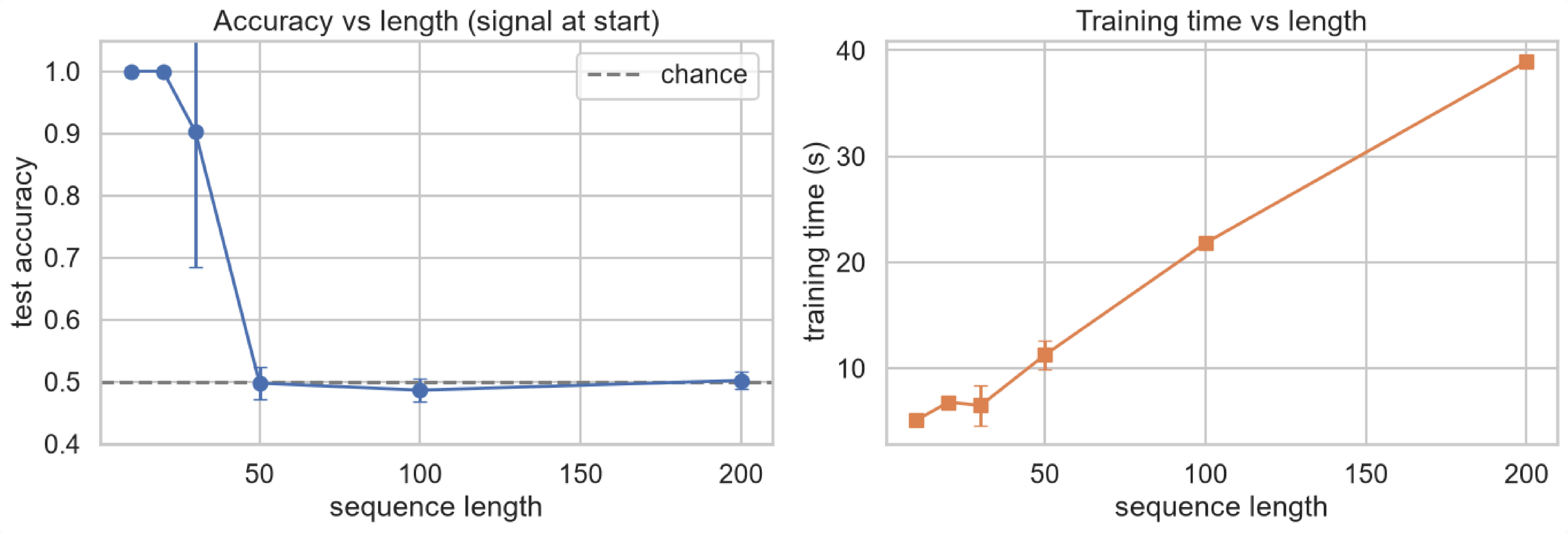
BASELINE

0.50

Random chance (binary task)

Perfect memory to L=20, collapse at L=30, chance at L=50

Accuracy and training time vs. sequence length



LENGTH	ACCURACY	SD
L = 10	1.000	-
L = 20	1.000	-
L = 30	0.900	±0.22
L = 50+	~0.50	-

TRAINING TIME

5.1s (L=10) to 38.9s (L=200)

7.6× on CPU - linear scaling

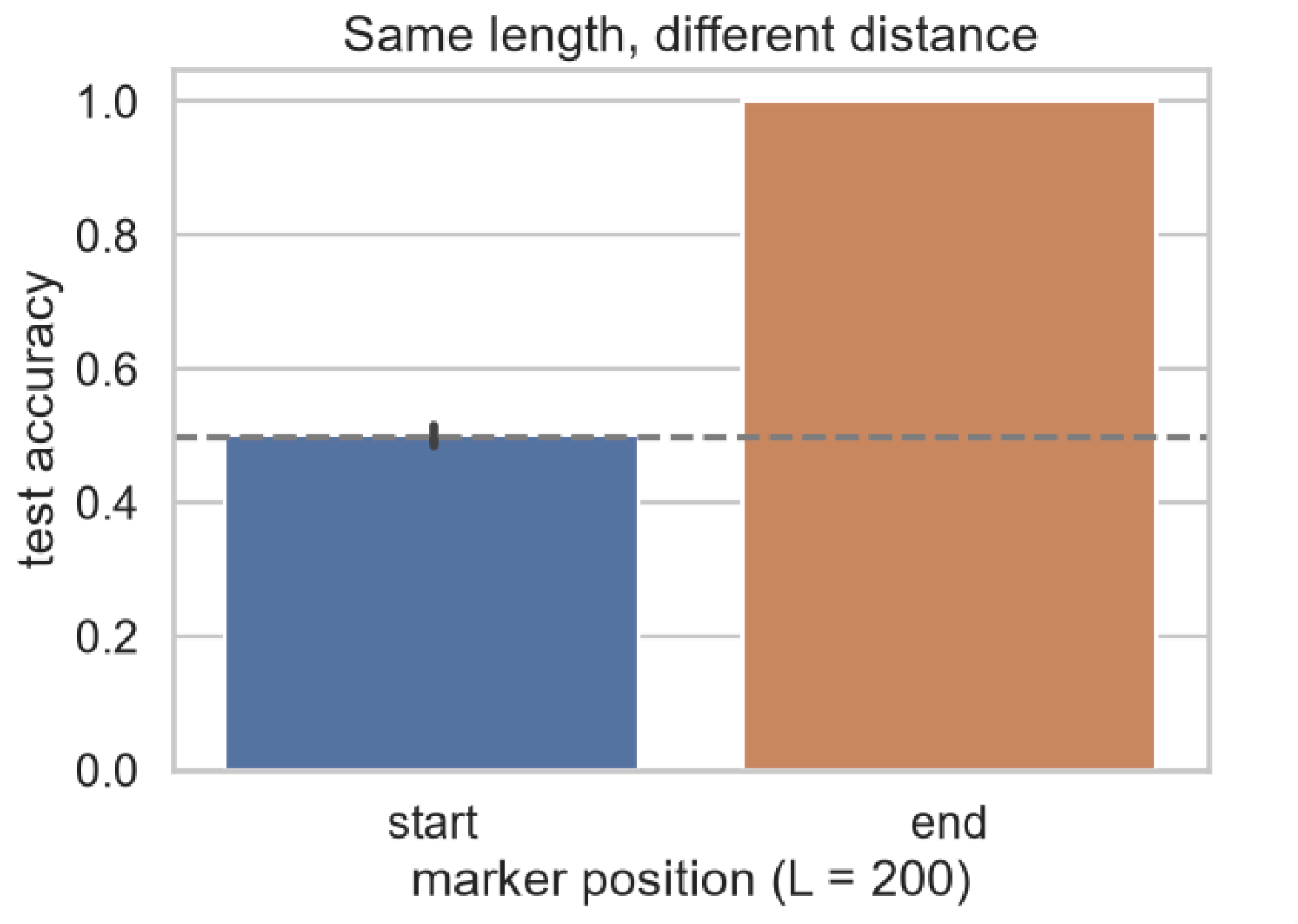
Same length, same compute - only distance differs

FIXED

L=200

Sequence length held constant for both conditions

Accuracy vs. marker-to-readout distance (L=200)



MARKER AT START

0.502

Distance = 200 steps · effectively chance

MARKER AT END

1.000

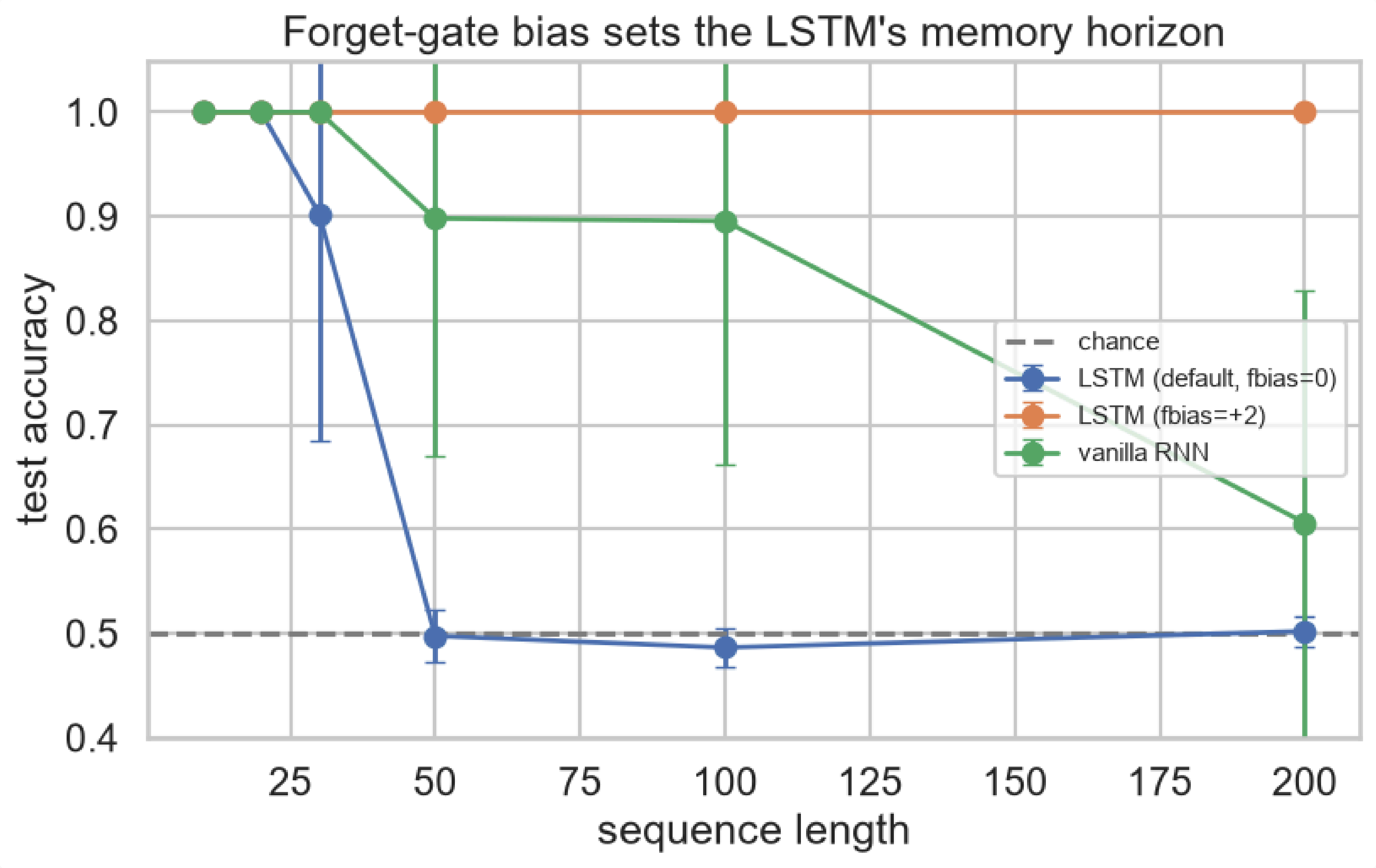
Distance ~0 steps · perfect recall

HEADLINE

Sequence-length sensitivity is dependency-distance sensitivity - length is a proxy, not the cause

PyTorch's default forget-gate bias was the problem all along

3-model accuracy comparison across sequence lengths



DEFAULT LSTM (PYTORCH)

`sigmoid(0) = 0.50`

Cell state halves every step - effective memory horizon ~30 steps

FORGET BIAS +2

`sigmoid(+2) = 0.88`

Near-full retention per step - perfect accuracy at every length up to 200

Jozefowicz et al. 2015 - recommended default: forget bias = 1

Architecture is rarely the binding constraint

TASK 1 · FAKE NEWS

From-scratch MLP = BiLSTM. Gains come elsewhere.

- Text-only architectures are statistically tied ($p = 0.69$)
- Pretraining (DistilBERT) gives the only real F1 lift on text
- Speaker metadata alone outperforms all text models

TASK 2 · LSTM MEMORY

LSTM horizon was an init artifact. Fix the forget gate.

- Sharp collapse at L=30 is not a fundamental LSTM limit
- Root cause: PyTorch default forget bias sigmoid(0)=0.5
- Forget bias +2 to perfect accuracy at all lengths up to 200

SHARED LESSON Understand the failure mode before reaching for a bigger model.